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Dung treated by high-temperature composting is an optimal bedding material for suckling calves according to analyses of microbial composition, growth performance, health status, and behavior

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ABSTRACT

Bedding materials are important for suckling buffalo calves. Treated dung has been used as a bedding material for dairy cows but the lack of an appropriate safety assessment limits its application. In this study, we evaluated the feasibility of treated dung (TD) as a bedding material for suckling calves by comparing TD with rice husk (RH) and rice straw (RS) bedding materials. The TD was prepared through high-temperature composting by *Bacillus subtilis*. Thirty-three newborn suckling buffalo calves (*Bubalus bubalis*, 40.06 ± 5.79 kg) were randomly divided into 3 bedding material groups (TD, RH, and RS) and bedded with 1 of the 3 bedding materials for 60 d. We compared cost, moisture content, bacterial counts, and microbial composition of the 3 bedding materials, and investigated growth performance, health status, behavior, rumen fermentation, and blood parameters of bedded calves. The results showed that TD contained the fewest gram-negative bacteria and coliforms on d 1 and 30 and the lowest relative abundance of *Staphylococcus* throughout the experiment. The RH and TD bedding materials had the lowest cost. Calves in the TD and RS groups showed a higher dry matter intake, and final body weight and average daily gain in the TD and RS groups tended to be higher than in the RH group. Calves in the TD and RS groups had a lower disease incidence (diarrhea and fever), fewer antibiotic treatments, and lower fecal score than calves in the RH group. Higher contents of IgG, IgA, and IgM were observed in calves of the TD and RS groups than in calves of the RH group on d 10, indicating higher immune ability in TD and RS groups. Furthermore, TD bedding increased the butyric acid content in the calf's rumen, whereas RS bedding increased the acetate content, which might be attributed to the longer time and higher frequency of eating bedding material in the RS group. Considering all of the above indicators, we concluded that TD is the optimal bedding material for calves based on economics, bacterial count, microbial diversity, growth performance, and health status. Our findings provide a valuable reference for bedding material choice and calf farming.

Key words: bedding material, suckling buffalo calf, microbial diversity, growth performance, health

INTRODUCTION

In modern dairy production, dairy cattle housing has evolved from tie-stall or stanchion barns, freestall mattress bedding barns, and compost bedded-pack barns (Bewley et al., 2017) to freestall harmless-recycled manure bedding barn. Bedding materials have changed along with the development of housing. In addition to traditional materials like concrete, sawdust, straw, and sand, the use of recycled manure solids (**RMS**), fermented manure, and treated dung (**TD**) has increased in recent years (Ferraz et al., 2020; Niu et al., 2022b).

Calves are the future of a farm. The calving period is the fastest growing period of a cow's life, the management of which affects reproduction, calving months, and lactation (Soberon and Amburgh, 2013; Krpálková et al., 2014). Bedding materials are closely related to growth performance, behavior, and health of calves (Panivivat et al., 2004). Bedding materials can influence the ADG of preweaning calves, and the ADG of animals bedded on sand or concrete is lower than that of animals bedded on straw, hay, and shavings (Shivley

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et al., 2018). Compared with sand, long straw as a bedding material can result in higher growth performance of calves (Hill et al., 2011). Bedding materials affect the daily distribution behavior pattern and lying time of calves, and hard bedding materials such as concrete reduce lying time and change the diurnal rhythm of calves (Hänninen et al., 2005; Ninomiya and Sato, 2009; Camiloti et al., 2012). Low-quality bedding materials pose stress on calves and produce pathogenic microbes that can harm calf health. Wet and hot bedding aggravates the heat stress on calves (Kovács et al., 2020). It has been reported that the number of *Escherichia coli* in rice husk bedding and straw bedding was greater than that on granite fines, sand, and wood shavings (Panivivat et al., 2004). The eggs of flies (*Musca domestica* L. and *Stomoxys calcitrans*) are less common in pine shavings than in sand, gravel, or sawdust bedding materials (Schmidtmann, 1991). Temperature and pH also affect bacterial reproduction (Ward et al., 2001), and acidified bedding can reduce the numbers of bacteria and fly eggs (*M. domestica*; Calvo et al., 2010).

With the increasingly limited resources in the future, manure management will become important, and recycling resources such as dung has potential. As the cost of bedding materials increases, a growing number of farms tend to use TD as bedding to improve the performance of dairy cows (Husfeldt and Endres, 2012). This material exhibits multiple advantages for cows such as increased comfort, reduced hock lesions, and improved cleanliness of dairy cows (Bradley et al., 2018). Although TD is a popular bedding material because of its economic, accessible, and recyclable characteristics, safety limits its application (Leach et al., 2015). Research has revealed that RMS bedding poses a higher risk to cows than straw bedding because of its microbial composition (Beauchemin et al., 2022). However, bacterial counts in RMS bedding are similar to those in many other bedding materials, and there have been no reports that RMS use is directly related to the clinical incidence of infectious diseases (Leach et al., 2015). As a result of these controversial results, farmers are cautious about the use of RMS for housing calves (Leach et al., 2015). Treated dung is achieved through high-temperature composting fermentation by *Bacillus subtilis* with antimicrobial activity. First, the dung is collected in the treatment area and the moisture content adjusted to about 60% by solid–liquid separation equipment or by adding auxiliary materials. Second, aerobic fermentation bacteria are added and the material mixture is regularly turned over until the temperature increases above 55°C. Treated dung can be rendered harmless if the fermentation temperature exceeds 55°C for 5 to 7 d during the treatment process (Niu et al., 2022a). Treated dung has a longer high-temperature fermentation period than RMS without added aerobic fermentation bacteria (Duan et al., 2020; Yang et al., 2021), thus reducing harmful bacteria.

Based on the above findings, we assumed that TD might be a feasible fermentation bedding with potential applications for calf raising. To our knowledge, no work has systematically evaluated the feasibility of TD as a bedding material for calves. Considering this, the objective of this study was to evaluate the feasibility and priority of TD as bedding material for suckling calves by comparing TD with rice husk (**RH**) and rice straw (**RS**) bedding. We hypothesized that TD would have a safe microbial composition and positive influence on the growth, health, and behavior of calves. We evaluated the economics, bacterial counts, and microbial composition of TD as a bedding material, as well as the effects on calves. This study provides a new strategy for improving the growth and health of calves by recycling dung to use as bedding material.

MATERIALS AND METHODS

Animal Management

All animal experiments in this study were approved by the Laboratory Animal Use and Care Committee in Hubei Science and Technology Agency (Hubei, China). The animal experiments followed the guidelines of Chinese national regulations regarding animal welfare ethics.

The experiments were conducted at Jinniu Animal Husbandry Co. Ltd. (Hubei, China). We selected 33 newborn Mediterranean buffalo calves (BW 40.06 ± 5.79 kg). Multiparous (parity 2 to 4) buffalo dams were artificially inseminated, and frozen semen (BP107240) was obtained from Inner Mongolia Saikexing Reproductive Biotechnology Co. Ltd. Calves were fed with 2 kg of colostrum within 2 h after birth, and then 5 kg of whole milk every day from d 1 to 7, 4 kg of milk from d 8 to 56, and 2 kg of milk from d 57 to 60. All calves were fed with milk twice a day at 0600 and 1700 h. During the experiment, the calves were allowed ad libitum access to clean water, starter feed (purchased from Huludao Meigao Animal Husbandry Co. Ltd.), and green hay (corn straw) from d 3 of life. The nutrients (DM basis) of the starter feed contained 20.72% CP, 3.27% ether extract, 23.16% NDF, 6.05% ADF, and 7.64% ash. The nutrient composition (DM basis) of the whole milk was 5.04% milk protein, 6.95% milk fat, 4.37% lactose, and 17.33% total solids. All calves were fed in individual pens (4.5 × 2.0 × 3.5 m) with individual feed and drinking barrels (20.5 cm in diameter, 18.0 cm deep,

5.9 L volume). To keep the animals clean and dry, feces were removed twice a week. A professional veterinarian measured rectal temperature and evaluated diarrhea of the calves at 0800 and 1600 h every day. If the rectal temperature was ≥39°C, the calf was diagnosed as having a fever and was treated with 10 mL of gentamicin sulfate and 4,000,000 units of benzylpenicillin. If the fecal score was ≥2, the calf was diagnosed as having diarrhea and was intravenously infused with a mixture of 250 mL of physiological saline, 250 mL of glucose, and 10 mL of gentamicin sulfate, and intramuscularly injected with 8 mL of berberine sulfate (Berchtold, 2009). All treatments lasted 1 to 3 d.

Experimental Design

We followed a randomized design, and groups were balanced by sex. The 33 calves were randomly divided into 3 groups (11 per group) and bedded on TD (6 male and 5 female), RH (5 male and 6 female), or RS (5 male and 6 female) for 60 d. Each calf was treated as an experimental unit. The thickness of bedding was 20 cm for each group.

Bedding Materials and Cost

Treated dung was prepared using a high-temperature composting fermentation method with a bacterial agent containing *Bacillus subtilis* DK068, lactic acid bacteria, yeast MX0126, cellulase, and lignin enzyme, with a viable bacteria count $>1.0 \times 10^{10}$ cfu/g, as previously reported (Niu et al., 2022b). The RH and RS materials were collected from a local farm. The properties of the 3 bedding materials, including the moisture content (**MC**), bulk density (**BD**), and particle size, were measured (Ferraz et al., 2020). For TD, the MC was 10.5 ± 0.2%, BD was 408.6 ± 9.6 kg/m^3, and 60% of the particles were less than <2 mm. For RH, MC was 11.3 ± 2.9%, BD was 117.6 ± 4.5 kg/m^3, and 90% of the particles measured 2 to 8 mm. For RS, MC was 13.2 ± 1.6%, BD was 68.9 ± 2.0 kg/m^3, and more than 90% of RS particles were >50 mm.

The raw material cost (US$/kg) of RH and RS was calculated based on material prices of the local farm over 4 mo (March, June, September, and December). The raw material cost of TD consisted 2 components: RH and the bacterial agent; the dung was obtained free from the farm. One kilogram of TD contained 0.2 kg of RH and 0.2 g of bacterial agent, and the price of the bacterial agent was based on a previous report (Niu et al., 2022a). All monetary values are presented in US dollars. The bedding material cost ($/m^2), based on a 0.2-m thickness, was calculated as follows: Bedding material cost = 0.2 × BD × raw material cost.

Growth and Feed Intake Data

Body weight was measured on d 1 (initial) and d 60 (final) using an electronic scale (ZF8003-T2, Kunshan Jiefeiyuan Measuring Equipment Co. Ltd.). Average daily gain was calculated as the difference between the final and initial BW divided by 60. Body size was measured on d 3 (initial) and d 60 (final), including withers height (vertical height from base of the front feet to the withers), body length (linear distance between the points of shoulder and rump), heart girth (circumference of the chest), and cannon circumference (circumference of thinnest tibia of left forelimb; Nemati et al., 2016).

The offers and leftovers were recorded to calculate DMI (including milk, starter feed, and green hay). Feed efficiency was calculated as BW gain (kg)/total DMI (kg).

Behavior Data

An infrared camera was used to record the behavior of the calves. Every 4 pens were equipped with 1 camera to monitor behavior. The behaviors of calves on d 15, 30, and 45 were recorded. The observation period was 12 h from 0900 to 1500 h and from 2100 to 0300 h every day to avoid the influence of human activities (feeding, treating, and sampling) on calf behaviors and to increase observation accuracy. The observation of behaviors was performed as previously reported (Wang et al., 2022), and the following indicators were recorded. Lying time referred to the time when a calf lay on the bedding in any position. Eating time represented the time from when a calf put its head into the starter feed barrel until the calf withdrew it. Eating frequency referred to the number of times a calf put its head into the starter feed barrel for more than 1 min. Eating bedding time referred to the time when the a calf's mouth was close to the bedding materials and the calf raised its head slowly while chewing. Eating bedding frequency referred to the number of times a calf ate bedding material for more than 1 min. Ruminating time referred to the time when calf chewed with any feed or bedding material taken in. Non-nutritional oral behavior was recorded when the mouth of calf licked the railing, barrel, or wall.

Calf Health Data

The rectal temperature of each calf was measured every day. If the temperature was ≥39°C, a calf was diagnosed as having a fever. The incidence of fever was calculated as follows: Incidence of fever (%) = number of calves with fever × fever days/(total number of calves × experiment days) × 100%.

Feces was scored every day using a previously reported method, where 0 = shaped and not hard, 1 = soft and slightly diffused, 2 = unshaped and diffused, and 3 = watery and mucous (McGuirk, 2008). If the fecal score of a calf was ≥2, the calf was diagnosed as having diarrhea. The incidence of diarrhea was calculated as follows: Incidence of diarrhea (%) = number of diarrhea calves × diarrhea days/(total number of calves × experiment days) × 100%.

Rumen Fluid Sampling and Parameter Measurements

Rumen fluid samples were collected from 5 healthy calves of similar weight in each group through the oral cavity using an oral stomach tube (MDW16, Chengdu Huazhi Kaiwu Co. Ltd.), as described by Shen et al. (2012), at 4 h after the morning feeding on d 60. The first 50-mL sample was discarded to avoid saliva contamination. After the rumen fluid was collected, the pH was immediately measured using a pH meter (FE-20-Five Easy Plus, Mettler Toledo Instruments Co. Ltd.). Then, the collected rumen fluid was filtered with a sterile 4-layer gauze. Afterward, the filtrate was centrifuged at 6,000 × *g* for 10 min at 4°C to collect the supernatant, and part of the supernatant was put into 5-mL centrifuge tubes. Subsequently, 5 mL of supernatant was mixed with 0.5 mL of metaphosphoric acid (25%) for VFA detection, and another 5 mL of supernatant was stored at −20°C for later ammonia-N measurement. Gas chromatography (GCMS isq LT, Thermo Fisher Scientific) was used for VFA detection (Hu et al., 2006), and phenol-sodium hypochlorite colorimetry was used for ammonia-N measurement (Weatherburn, 1967).

Blood Sampling and Parameter Measurements

Blood samples were collected through the jugular vein using vacuum blood collection tubes (5 mL containing 12–30 IU/mL heparin sodium; Winner Medical Co. Ltd.) at 0800 h on d 10 and 60. Plasma was obtained by centrifugation at 1,500 × *g* for 10 min at 25°C and stored at −20°C. We measured blood parameters related to metabolites, including plasma total protein (**TP**), total cholesterol (**TC**), glucose (**Glu**), BUN, creatine (**CRE**), and BHB. We also measured the growth-related parameter growth hormone (**GH**). We measured immunity-related parameters including IgA, IgG, IgM, IL-2, and IL-1β. Total protein was detected by Coomassie Brilliant Blue method; TC was determined by cholesterol oxidase phenol 4-aminoantipyrine peroxidase method; Glu was measured by glucose oxidase method; BUN was measured by urease method; and CRE was detected by sarcosine oxidase method. Total protein, TC, Glu, BUN, and CRE were determined by assay kits (Nanjing Jiancheng Bioengineering Institute); GH, BHB, IgA, IgG, IgM, IL-2, and IL-1β were detected by ELISA using kits (Quanzhou Ruixin Biotechnology Co. Ltd.). All measurements were performed according to previously reported methods (Li et al., 2014).

Bedding Material Sampling and Parameter Measurements

Three replicate samples of bedding materials were collected from each group by the 5-spot method on d 1, 15, 30, 45, and 60. Samples collected on d 1, 30, and 60 were put into 2 sterile centrifuge tubes, with one used for bacterial count and the other stored at −20°C for microbioassay. All bedding material samples were dried in the oven (Jinghong Experimental Equipment Co. Ltd.) at 65°C for 24 h for moisture content measurement.

After being cultured, bacteria were counted (Rowbotham and Ruegg, 2016). Ten grams of fresh bedding material sample was put into 90 mL of sterile PBS in a resealable plastic bag, and mixed using a vortex oscillator (MX-S, Scilogex LLC) at low speed for 1 min. Then, the PBS mixture was subjected to serial dilution at a diluent:PBS ratio of 1:9. The diluted PBS at a gradient concentration from 1×10^3 to 1×10^7 was coated on the surface of agar plates in triplicate. MacConkey agar (Qingdao Hope Bio-Technology Co. Ltd.) was used to identify gram-negative bacteria and coliforms. Agar plates were incubated for 24 h at 37°C. All colonies were defined as gram-negative bacteria, and lactose-positive colonies were defined as coliforms. Bacterial counts on the plates with 30 to 300 colonies were expressed in colony-forming units per gram, which were converted to log colony-forming units per gram.

Microbial DNA was extracted from bedding materials following the instructions of the TGuide S96 Magnetic Soil/Stool DNA Kit (Tiangen Biochemical Technology Co. Ltd.). The DNA concentration was measured using the Qubit dsDNA HS Assay Kit and Qubit 4.0 Fluorometer (Thermo Fisher Scientific). PCR amplification of the V3-V4 region of 16S rRNA gene was performed using forward primer 338F (5′-ACTCCTACGGGAGGCAGCA-3′) and reverse primer 806R (5′-GGACTACHVGGGTWTCTAAT-3′). The 10-μL PCR system contained 50 ng of DNA template, 0.3 μL of each primer, 5 μL of KOD FX Neo Buffer, 2 μL of dNTP (2 m*M* each), 0.2 μL of KOD FX Neo, and 2.2 μL of deionized distilled H_2O. The PCR conditions were as follows: predenaturation at 95°C for 5 min, followed by 25 cycles at 95°C for 30 s, 50°C for 30 s, and 72°C for

40 s, and, finally, extension at 72°C for 7 min. The PCR products were purified with Agencourt AMPure XP beads (Beckman Coulter) and quantified following the instructions of Qubit dsDNA HS Assay Kit and Qubit 4.0 Fluorometer. After library construction of PCR products, the Novaseq 6000 (Illumina) was used for sequencing of 250-bp paired-end reads. All DNA extraction and sequencing of bedding materials were conducted by Biomarker (Beijing, China).

The raw reads of sequencing data were primarily filtered according to quality of single nucleotide by Trimmomatic (version 0.33; Edgar, 2013). Then, primer sequences were identified and removed by Cutadapt (version 1.9.1) to obtain clean reads. USEARCH (version 10) was used to assemble the clean reads, and then UCHIME (version 8.1) was used to remove chimera to obtain effective reads. The effective reads were clustered into the same operational taxonomic units (OTU) at a level of similarity of ≥97% by USEARCH (version 10). Taxonomic assignment of the OTUs was performed using QIIME2 according to the SILVA database (http://www.arb-silva.de) with a confidence threshold of 70%.

Statistical Analyses

Data were analyzed using SPSS version 21.0 (IBM Corp.). After normality and homogeneity tests, one-way ANOVA was performed to compare differences in rumen fluid, antibiotic treatment times, and blood parameters between groups.

A linear regression method was performed to analyze the correlation between final body size and the covariance of initial body size. Final body size was analyzed using covariance analysis as follows:

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \tau_i + \varepsilon_{ij},$$

where y_{ij} is the dependent variable of calf j in bedding treatment i, β_0 is the intercept, β_1 is the regression coefficient, x_{ij} is the covariate, τ_i is the fixed effect of bedding treatments, and ε_{ij} is the random error. Final BW, ADG, total DMI, and feed efficiency were also analyzed using above model with the covariance of initial BW.

Moisture content; bacterial count; milk, starter feed, and green hay intake data; and behavior data were analyzed using a statistical mixed model (repeated measures) as follows:

$$y_{ijk} = \mu + \tau_i + \delta_{ij} + t_k + (\tau \times t)_{ik} + \varepsilon_{ijk},$$

where y_{ijk} is the dependent variable of calf j in bedding treatment i at time k, μ is the overall mean, τ_i is the fixed effect of bedding treatments, δ_{ij} is the random effect of covariance between repeated measures in an individual, t_k is the fixed effect of measurement time, $(\tau \times t)_{ik}$ is the fixed effect of the interaction between bedding treatments and measurement time, and ε_{ijk} is the random error of the variance between the measures within the individual. Initial BW was included in the model as a covariate for milk, starter feed, and green hay intake data analyses.

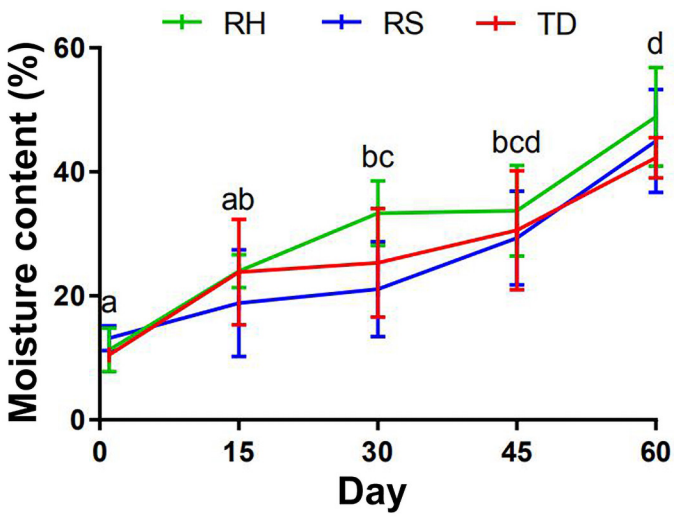


Figure 1. Moisture content of bedding materials over time. RH = rice husk; RS = rice straw; TD = treated dung. Different lowercase letters (a–d) indicate significant differences ($P < 0.05$). Error bars represent SEM.

The Kruskal-Wallis test was adopted for analysis of cost and fecal score. The Duncan method was used for pairwise comparison of significant differences. All data are expressed as means and standard error of mean. The incidences of diarrhea and fever were analyzed by chi-squared test, and results were expressed as means. $P < 0.05$ was considered to indicate a significant difference, and $0.05 \leq P < 0.10$ as a tendency.

RESULTS

Moisture Content

The MC of bedding materials over time is shown as mean and standard error of the mean in Figure 1. The MC of all 3 beddings was <15% on d 1 and did not differ significantly among the 3 materials throughout the 60-d experimental period. However, MC did differ significantly over time ($P < 0.01$).

Microorganisms in Bedding Materials

The results of bacterial counts are shown in Table 1. On d 1, there was a significant difference in the gram-negative bacterial count; TD had the fewest bacteria,

Table 1. Bacterial counts of the 3 bedding materials

Item	Bedding treatment[1]			SEM	*P*-value[2]		
	RH	RS	TD		B	T	B × T
Gram-negative bacteria, $\log_{10}$ cfu/g							
d 1	7.1[a]	6.6[b]	5.8[c]	0.20	<0.01	<0.01	0.01
d 30	7.2[a]	6.6[b]	6.2[c]	0.15			
d 60	7.8[a]	6.7[b]	6.8[b]	0.05			
Coliforms, $\log_{10}$ cfu/g							
d 1	5.7[a]	5.6[a]	4.7[b]	0.07	<0.01	<0.01	0.01
d 30	6.6[a]	6.0[b]	5.4[c]	0.10			
d 60	7.4[a]	6.5[b]	6.4[b]	0.17			

[a–c]Values with different lowercase letters in the same row are significantly different ($P < 0.05$).

[1]Bedding materials: RH = rice husk; RS = rice straw; TD = treated dung.

[2]Treatment effects: B = effect of bedding treatments; T = effect of measurement time; B × T = interaction between bedding treatments and measurement time.

whereas RH had the most ($P < 0.01$). The coliform count was significantly lower in TD than in RH and RS ($P < 0.01$). The gram-negative bacteria and coliform counts in bedding materials were different among the 3 groups on d 30: the TD group had a significantly lower count than the other 2 groups ($P < 0.01$). On d 60, the gram-negative bacteria and coliform counts were similar between TD and RS and lower than those in RH ($P < 0.01$).

The microorganisms at the phylum level in the 3 bedding materials during the experimental period are shown in Figure 2A. On d 1, *Proteobacteria* (28.2%) and *Chloroflexi* (20.9%) were the top 2 phyla in TD. Over time, the relative abundance of *Chloroflexi* declined and that of *Firmicutes* increased on d 30 and 60. In RH, the relative abundances of *Proteobacteria* (49.0%) and *Firmicutes* (27.1%) were highest, but that of *Proteobacteria* was reduced on d 30 and 60, whereas abundance of *Actinobacteria* was increased. On d 1, the relative abundance of *Proteobacteria* (52.4%) was highest in RS and it decreased on d 30 and 60, whereas that of *Firmicutes* increased over time.

Microorganisms at the genus level in the 3 types of bedding at different times are shown in Figure 2B. The results showed significant differences among the 3 beddings. On d 1, the top genus in TD was *Truepera* (5.4%), whereas *Staphylococcus* had the highest relative abundance (24.5%) in RH, and *Sphingobacterium* (12.6%) had the highest relative abundance in RS. On d 30, *Staphylococcus* exhibited the highest relative abundance of all 3 bedding materials. On d 60, *Acinetobacter* (3.9%) had the highest relative abundance

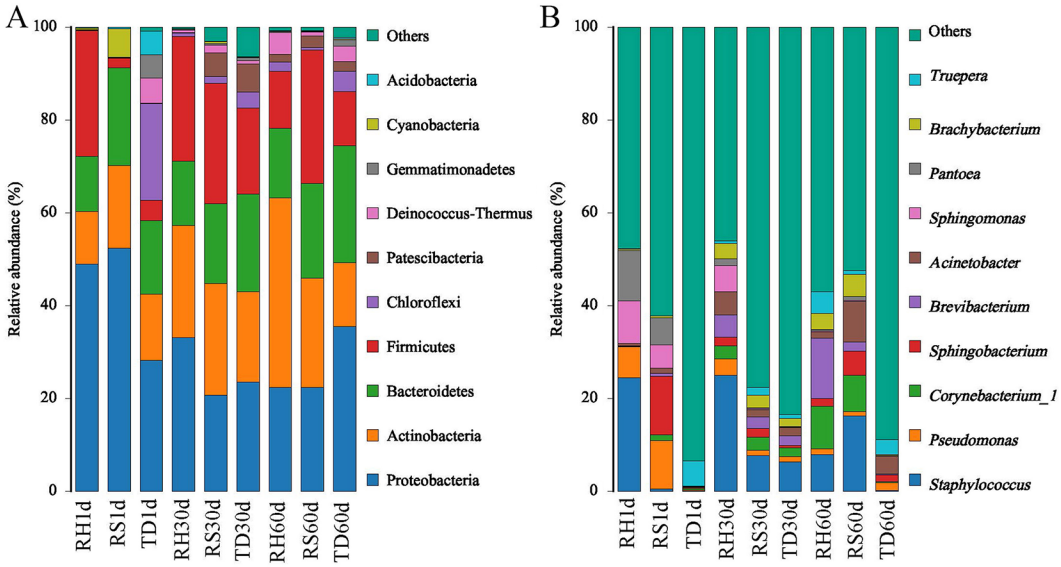


Figure 2. Microorganism distribution in bedding materials on d 1, 30, and 60 at the (A) phylum level and (B) genus level. RH = rice husk; RS = rice straw; TD = treated dung.

Table 2. Cost of the 3 bedding materials

Item	Bedding treatment[1]			SEM	*P*-value
	RH	RS	TD		
Raw material cost, $/kg	0.04[a]	0.10[b]	0.01[c]	0.01	<0.01
Bulk density, kg/m^3	117.6	68.9	408.6	—	—
Bedding material cost, $/m^2	1.0[a]	1.4[b]	0.9[a]	0.06	<0.01

[a–c]Values with different lowercase letters in the same row are significantly different ($P < 0.05$).

[1]Bedding materials: RH = rice husk; RS = rice straw; TD = treated dung.

[2]The bedding material cost was calculated based on 0.2-m thickness.

in TD; *Brevibacterium* (13.0%) was highest in RH, and *Staphylococcus* (16.2%) was highest in RS on 60 d.

Bedding Cost

Bedding cost is shown in Table 2. Treated dung had the lowest raw material cost ($0.01/kg); RH had a medium cost ($0.04/kg) and RS had the highest cost ($0.10/kg; $P < 0.01$). After the materials were bedded, the bedding material costs of RH ($1.00/m^2) and TD ($0.90/m^2) were significantly lower than that of RS ($1.40/m^2; $P < 0.01$).

Growth Performance and Feed Intake

The results of initial and final BW and body size are shown in Table 3. A tendency for a difference ($P = 0.09$) in final BW and ADG was observed among the 3 beddings. The final BW in RH calves was 72.4 kg, which was lower than that in TD (79.7 kg) and RS (79.8 kg) calves. The ADG in RH calves was 0.54 kg/d, which was lower than that in TD (0.66 kg/d) and RS (0.66 kg/d) calves. No significant difference in body size was found among the 3 bedding materials.

The results of milk, starter feed, and green hay intake, total DMI, and feed efficiency are shown in Table 4. From d 1 to 20, the daily starter feed intake in the TD group (21.3 g/d) was significantly higher than that in RH group (8.7 g/d; $P = 0.02$), and milk intake in the TD group (777.5 g/d) and RS group (780.4 g/d) was higher than that in the RH group (748.2 g/d; $P < 0.01$). From d 21 to 40, daily starter feed intake in the TD group (41.9 g/d) and RS group (30.9 g/d) tended to be higher than that in the RH group (20.6 g/d; $P = 0.07$). From d 41 to 60, calves in the RH group (58.5 g/d) had a lower daily starter feed intake than calves in the TD (188.3 g/d) or RS (157.1 g/d) groups ($P < 0.01$). However, a significant difference ($P < 0.01$) in total DMI was observed among 3 bedding groups: calves in the RH group (44.8 kg) had lower DMI than those in the TD group (48.7 kg) and RS group (48.1 kg). No significant difference in green hay intake or feed efficiency was observed.

Differences in GH and metabolites in calf blood from 3 bedding groups were apparent on d 10 and 60 (Table 5). On d 10, GH was significantly lower in the RH group (1.1 ng/mL) than in the TD group (1.3 ng/mL) or RS group (1.3 ng/mL; $P = 0.02$), whereas TC was significantly higher in the TD group (19.7 mmol/L) than in the RS group (13.5 mmol/L) or RH group (13.5 mmol/L; $P = 0.01$). On d 60, BUN was significantly higher in the TD group (2.5 mmol/L) than in the RH group (1.9 mmol/L) or RS group (1.6 mmol/L; $P = 0.03$).

Table 3. Body weight, ADG, and body size of suckling buffalo calves in 3 bedding material groups

Item	Bedding treatment[1]			SEM	*P*-value
	RH	RS	TD		
Initial BW (d 1), kg	38.9	40.8	40.6	1.06	0.73
Final BW (d 60), kg	72.4	79.8	79.7	2.62	0.09
ADG, kg/d	0.54	0.66	0.66	0.04	0.09
Initial withers height (d 3), cm	72.6	73.7	73.9	0.64	0.70
Final withers height (d 60), cm	83.6	85.0	85.9	1.57	0.60
Initial body length (d 3), cm	59.4	61.4	60.2	0.76	0.58
Final body length (d 60), cm	79.3	79.7	79.6	1.65	0.98
Initial heart girth (d 3), cm	77.8	79.3	79.4	0.73	0.69
Final heart girth (d 60), cm	99.8	102.0	103.4	2.30	0.66
Initial cannon circumference (d 3), cm	12.5	12.5	13.2	0.22	0.31
Final cannon circumference (d 60), cm	14.7	14.5	15.0	0.49	0.63

[1]Bedding materials in different groups: RH = rice husk; RS = rice straw; TD = treated dung.

Table 4. Starter feed and green hay intake, total DMI, and feed efficiency of suckling buffalo calves (DM basis) in different bedding groups

	Bedding treatment[1]				*P*-value[2]		
Item	RH	RS	TD	SEM	B	T	B × T
Starter (d 1–20), g/d	8.7[a]	19.7	21.3[b]	3.94	0.02	0.01	<0.01
Starter (d 21–40), g/d	20.6	30.9	41.9	4.96	0.07	<0.01	0.13
Starter (d 41–60), g/d	58.5[a]	157.1[b]	188.3[b]	14.1	<0.01	<0.01	<0.01
Green hay (d 1–20), g/d	2.3	3.5	3.4	0.49	0.23	<0.01	0.30
Green hay (d 21–40), g/d	6.7	8.0	6.8	2.64	0.51	<0.01	0.62
Green hay (d 41–60), g/d	31.2	40.7	33.8	7.26	0.44	<0.01	0.01
Milk (d 1–20), g/d	748.2[a]	780.4[b]	777.5[b]	56.94	<0.01	<0.01	0.92
Milk (d 21–40), g/d	709.3	713.4	715.4	4.83	0.66	0.25	0.94
Milk (d 41–60), g/d	656.5	654.3	656.1	1.38	0.49	<0.01	0.61
Total DMI, kg	44.8[a]	48.1[b]	48.7[b]	0.43	<0.01	—	—
Feed efficiency	0.74	0.83	0.82	0.06	0.50	—	—

[a,b]Values with different lowercase letters in the same row are significantly different ($P < 0.05$).

[1]Bedding materials: RH = rice husk; RS = rice straw; TD = treated dung.

[2]Treatment effects: B = effect of bedding treatments; T = effect of measurement time; B × T = interaction between bedding treatments and measurement time.

Health Status

Data on fecal score, disease incidence, and total antibiotic treatment times are shown in Table 6. From d 1 to 20, no significant differences were observed in fecal score or diarrhea incidence among 3 groups, whereas the incidence of fever was higher in the RH group (16.3%) than in the TD group (10.0%) and RS group (7.5%; $P = 0.06$). From d 21 to 40, fecal score was significantly higher in the RH group (1.3) than in the TD group (1.1) and RS group (1.1; $P < 0.01$), and the diarrhea incidence was significantly higher in the RH group (18.5%) than in the TD (4.8%) and RS (5.1%) groups ($P < 0.01$). Fever incidence was 22.7% in RH group, which was significantly higher than in the TD (7.7%) and RS (11.1%) groups ($P = 0.01$). The disease incidence and fecal score declined from d 41 to 60 in all 3 bedding groups, and there was no significant difference among the 3 groups. Overall, fecal score, diarrhea incidence, and fever incidence were significantly higher in the RH group than in the TD and RS groups ($P = 0.01$). Additionally, the total number of antibiotic treatments was significantly higher in the RH group (4.4) than in the TD (2.2) and RS (2.0) groups ($P = 0.03$).

Immunity-related serum parameters of suckling buffalo calves are shown in Table 7. Plasma immunity had a negative relation with the incidences of diarrhea and

Table 5. Growth hormone and metabolites of suckling buffalo calves in the 3 bedding material groups

	Bedding treatment[2]				
Item[1]	RH	RS	TD	SEM	*P*-value
d 10					
GH, ng/mL	1.1[b]	1.3[a]	1.3[a]	0.04	0.02
TP, g/L	47.5	43.8	48.9	1.74	0.71
TC, mmol/L	13.5[b]	13.5[b]	19.7[a]	1.02	0.01
Glu, mmol/L	9.2	8.3	9.5	0.48	0.95
BUN, mmol/L	2.2	2.2	2.3	0.13	0.95
CRE, μmol/L	145.3	131.1	145.8	10.6	0.82
BHB, mmol/L	0.8	0.8	0.8	0.03	0.57
d 60					
GH,ng/mL	1.0	1.2	1.2	0.05	0.47
TP, g/L	49.2	49.9	51.5	0.77	0.46
TC, mmol/L	22.4	25.1	28.9	2.23	0.53
Glu, mmol/L	7.3	8.0	6.4	0.54	0.49
BUN, mmol/L	1.9[b]	1.6[b]	2.5[a]	0.13	0.03
CRE, μmol/L	126.2	134.5	146.4	4.96	0.28
BHB, mmol/L	0.6	0.6	0.6	0.02	0.66

[a,b]Values with different lowercase letters in the same row are significantly different ($P < 0.05$).

[1]GH = growth hormone; TP = total protein; TC = total cholesterol; Glu = glucose; CRE = creatine.

[2]Bedding materials: RH = rice husk; RS = rice straw; TD = treated dung.

Table 6. Fecal score and incidence of diarrhea and fever of suckling buffalo calves on different bedding materials

Item	Bedding treatment[1]			SEM	*P*-value
	RH	RS	TD		
Fecal score[2] (d 1–20)	1.2	1.2	1.2	0.04	0.57
Fecal score (d 21–40)	1.3[a]	1.1[b]	1.1[b]	0.07	<0.01
Fecal score (d 41–60)	1.2	1.0	1.0	0.05	0.23
Fecal score (d 1–60)	1.3[a]	1.1[b]	1.1[b]	0.02	0.01
Incidence of diarrhea (d 1–20), %	14.4	9.2	13.3	—	0.41
Incidence of diarrhea (d 21–40), %	18.5[a]	5.1[b]	4.8[b]	—	<0.01
Incidence of diarrhea (d 41–60), %	2.2	0.0	0.0	—	0.26
Total incidence of diarrhea (d 1–60), %	11.7[a]	4.8[b]	6.0[b]	—	0.01
Incidence of fever (d 1–20), %	16.3	7.5	10.0	—	0.06
Incidence of fever (d 21–40), %	22.7[a]	11.1[b]	7.7[b]	—	0.01
Incidence of fever (d 41–60), %	0.0	0.0	0.0	—	1.00
Total incidence of fever (d 1–60), %	13.0[a]	6.2[b]	5.9[b]	—	0.01
Total treatment with antibiotics	4.4[a]	2.0[b]	2.2[b]	0.44	0.03

[a,b]Values with different lowercase letters in the same row are significantly different ($P < 0.05$).

[1]Bedding materials: RH = rice husk; RS = rice straw; TD = treated dung.

[2]Fecal score scale: 0 = shaped and not hard; 1 = soft and slightly diffused; 2 = unshaped and diffused; 3 = watery and mucous. Fecal score ≥2 was identified as diarrhea.

fever. On d 10, the IgA content was significantly lower in the RH group (13.7 μg/mL) compared with the TD and RS groups (35.8 and 36.0 μg/mL, respectively; $P = 0.04$). The same trend was observed for IgM (12.6 μg/mL in RH vs. 56.7 μg/mL in TD and 62.09 μg/mL in RS; $P = 0.01$). However, IgG content was significantly lower in RH (421.1 μg/mL) than in TD (661.7 μg/mL; $P = 0.04$). No significant differences in the contents of IL-2 and IL-1β were observed among the 3 groups. On d 60, there was no difference among groups in all tested parameters.

Behavior

The behaviors of suckling buffalo calves are listed in Table 8. We observed no significant differences in lying time, eating time, eating frequency, ruminating time, or non-nutritional oral behavior time among the 3 bedding groups. However, significant differences in the time and frequency of bedding material eating were observed among the 3 groups ($P < 0.01$). On d 15, calves in the TD group exhibited a significantly lower duration and frequency of bedding material eating than calves in the RH and RS groups ($P < 0.01$). On d 30, the bedding material eating duration and frequency were significantly lower in the TD and RH groups than in the RS group. On d 45, there was a significant difference in bedding material eating duration and frequency among the 3 groups, with calves in the TD group exhibiting the lowest values and those in the RS group the highest. Interestingly, bedding material eating duration and frequency in TD and RH declined over time, whereas the same measures increased in the RS group.

Rumen Fermentation

As shown in Table 9, pH and ammonia-N did not differ significantly among the 3 bedding groups. The acetic acid content was lower in the TD group (43.1%) than in the RS group (49.6%; $P = 0.04$). However, calves in the TD group had a higher butyric acid content than calves in the RH and RS groups (13.2% vs. 9.1% and 9.1%; $P = 0.04$).

Table 7. Serum parameters related to immune status of suckling buffalo calves in different bedding material groups

Item	Bedding treatment[1]			SEM	*P*-value
	RH	RS	TD		
d 10					
IgA, μg/mL	13.7[a]	36.0[b]	35.8[b]	4.53	0.04
IgG, μg/mL	421.1[a]	599.2	661.7[b]	46.4	0.04
IgM, μg/mL	12.6[a]	62.1[b]	56.7[b]	7.59	0.01
IL-2, pg/mL	26.9	30.1	30.1	1.01	0.33
IL-1β, pg/mL	15.9	17.4	16.6	0.69	0.67
d 60					
IgA, μg/mL	77.5	69.7	78.2	9.89	0.93
IgG, μg/mL	877.3	916.2	939.6	57.0	0.92
IgM, μg/mL	144.7	132.3	179.1	20.0	0.65
IL-2, pg/mL	42.1	44.4	39.4	1.72	0.52
IL-1β, pg/mL	18.5	18.7	17.5	0.54	0.64

[a,b]Values with different lowercase letters in the same row are significantly different ($P < 0.05$).

[1]Bedding materials: RH = rice husk; RS = rice straw; TD = treated dung.

Table 8. Behavior characteristics of suckling buffalo calves in different bedding material groups (within 12 h)

Item	Bedding treatment[1]			SEM	*P*-value[2]		
	RH	RS	TD		B	T	B × T
Lying time (d 15), min	561	551	541	7.64	0.26	0.01	0.07
Lying time (d 30), min	518	561	542	8.90			
Lying time (d 45), min	515	546	517	7.25			
Eating time (d 15), min	18	16	24	1.99	0.71	<0.01	0.66
Eating time (d 30), min	22	30	31	3.56			
Eating time (d 45), min	41	38	40	3.51			
Eating frequency (d 15), n	3.5	3.6	5.3	0.50	0.39	0.09	0.59
Eating frequency (d 30), n	4.3	4.2	4.9	0.47			
Eating frequency (d 45), n	5.1	6.6	5.6	0.55			
Bedding material eating time (d 15), min	28[a]	31[a]	9[b]	2.98	<0.01	0.32	0.03
Bedding material eating time (d 30), min	23[a]	44[b]	8[a]	4.05			
Bedding material eating time (d 45), min	18[a]	41[b]	3[c]	3.11			
Bedding material eating frequency (d 15), n	5.6[a]	6.1[a]	1.9[b]	0.59	<0.01	0.15	0.01
Bedding material eating frequency (d 30), n	4.5[a]	8.8[b]	1.7[a]	0.81			
Bedding material eating frequency (d 45), n	2.9[a]	8.2[b]	0.6[c]	0.61			
Ruminating (d 30), min	27	37	26	1.70	0.12	<0.01	0.95
Ruminating (d 45), min	58	66	58	4.22			
NNOB[3] (d 15), min	41	48	61	3.92	0.15	0.01	0.40
NNOB (d 30), min	41	39	48	3.72			
NNOB (d 45), min	30	34	30	1.83			

[a–c]Values with different lowercase letters in the same row are significantly different ($P < 0.05$).

[1]Bedding materials: RH = rice husk; RS = rice straw; TD = treated dung.

[2]Treatment effects: B = effect of bedding treatments; T = effect of measurement time; B × T = interaction between bedding treatments and measurement time.

[3]NNOB = non-nutritional oral behavior.

DISCUSSION

Proteobacteria, *Firmicutes*, *Bacteroidetes*, and *Chloroflexi* are the dominant phyla in the high-temperature fermentation process of manure; *Chloroflexi* in particular can tolerate elevated temperatures (Meng et al., 2019). In this study, the relative abundance of *Chloroflexi* was higher in TD than in RH and RS due to the unique high-temperature fermentation period of TD; this result is consistent with a previous report (de Hoon et al., 2010). The main function of *Firmicutes* colonizing the animal gut is to utilize various carbohydrates, and most of them have spores, which can extend their survival time (Chen et al., 2017). On d 30 and 60, the relative abundance of *Chloroflexi* decreased but that of *Firmicutes* increased with the feces excreted by calves. Our data showed that in terms of relative abundance, *Actinobacteria* ranked second in the RH group, whereas *Bacteroidetes* ranked second in the RS group. *Actinobacteria* exist widely in soil, and *Bacteroidetes* mainly colonize the animal gut. Based on this, we speculated that the microbial compositions of RH and RS might be closely related to the environment, and thus they might be uncontrollable. At the genus level, *Truepera* is mainly involved in the mineralization of OM, nitrogen cycle, utilization, and transformation (Zhang et al., 2020). Our data showed that in TD, heat-tolerant *Truepera* exhibited the highest relative abundance, which might be due to the high-temperature fermentation used to prepare TD, and because the relative abundance of heat-labile pathogenic bacteria was significantly decreased, which was supported by bacterial count results showing that gram-negative bacteria and coliform counts were lowest in TD on d 1 and 30. Other bacteria such as *Staphylococcus* and *Acinetobacter* gradually colonized TD on d 30 and 60. *Staphylococcus* is the most important pathogenic

Table 9. Rumen fermentation of suckling buffalo calves in different bedding material groups

Item	Bedding treatment[1]			SEM	*P*-value
	RH	RS	TD		
pH	6.4	6.5	6.5	0.08	0.76
Ammonia-N, mg/dL	14.7	19.4	18.4	2.03	0.68
Total VFA, mmol/L	33.9	37.5	35.1	2.19	0.92
Acetic acid, %	46.6	49.6[a]	43.1[b]	1.24	0.04
Propionic acid, %	33.4	30.0	29.6	0.97	0.26
Butyric acid, %	9.1[a]	9.1[a]	13.2[b]	0.78	0.04
Isobutyric acid, %	5.0	5.0	6.0	0.35	0.47
Valeric acid, %	2.9	2.5	4.4	0.35	0.20
Isovaleric acid, %	3.1	3.7	3.7	0.33	0.63
A:P[2]	1.4	1.6	1.6	0.08	0.47

[a,b]Values with different lowercase letters in the same row are significantly different ($P < 0.05$).

[1]Bedding materials: RH = rice husk; RS = rice straw; TD = treated dung.

[2]A:P = acetic/propionic.

bacteria causing inflammation (Adkins et al., 2022). In this study, *Staphylococcus* exhibited a high relative abundance in RH on d 1 and 30 but decreased by d 60, whereas *Staphylococcus* had a low abundance in RS on d 1 and increased over time. One possible reason might be that RS was an excellent medium for *Staphylococcus*, and this opposite trend might be related to the properties of different beddings, such as pH and C:N ratio. Overall, the microbial compositions of RH and RS were complex and environment-dependent. In constrast, TD exhibited relatively simple microbial compositions and a lower level of pathogenic bacterial abundance after its high-temperature fermentation period. Therefore, TD was a safer bedding material in terms of microbial composition and bacterial counts.

Diarrhea is a common disease in calves and the main cause of death (Renaud et al., 2020). Diarrhea is harmful to the growth of calves, damages their immunity, and is often accompanied by a reduction in ADG (Donovan et al., 1998; Graham et al., 2018). Previous studies have revealed that *Escherichia coli* and *Salmonella* are common bacteria causing calf diarrhea (Cho and Yoon, 2014). Consistently, our results demonstrated that the incidence of diarrhea and fecal score were positively related to counts of gram-negative bacteria and coliforms. Based on these findings, we speculated that the gram-negative bacteria and coliforms might cause diarrhea of calves in this study, which remains to be confirmed. Immunoglobulins regulating immune function are important to protect newborn calves from infection. The immune system of calves is weakest before the age of 30 d, when calves are in a "window of susceptibility." In this period, their immune system is immature, and the immunity acquired obtained from colostrum is gradually declining (Chase et al., 2008). Immunoglobulin A, IgG, and IgM play valuable roles in humoral immunity. Calves with low-level IgG have been reported to exhibit a higher incidence rate of diarrhea than calves with normal levels of IgG (Wells et al., 1996). In this study, the contents of IgA, IgG, and IgM were higher in TD and RS groups compared with RH group on 10 d, which might explain the lower incidence of diarrhea in these 2 bedding groups. The incidences of diarrhea and fever decreased after d 41 due to maturation of the immune system and treatment by the veterinarian, and immunoglobulin levels returned to normal on d 60. Interleukin-2 is a cytokine with strong proinflammatory activity (Kosmaczewska, 2014), and IL-1β is an important mediator in the inflammatory response (Ma et al., 2021). In this study, on d 10 and 60, there was no difference in IL-2 and IL-1β levels among the 3 bedding groups. Overall, TD and RS as bedding materials were associated with a lower incidence of diarrhea and fever and a lower fecal score and with higher immunity, which we speculated might be related to the bacterial counts in the bedding material.

The growth performance of calves has a direct effect on adult production. Previous research has indicated that a high incidence of disease significantly reduces ADG in preweaning dairy heifers (Shivley et al., 2018). Our results showed that the lower DMI in the RH group resulted in a tendency toward lower final BW and ADG. We speculated that the lower DMI in the RH group might be related to their higher disease incidence because fever would result in loss of appetite. Our speculation was confirmed by the results that TC was lower in the RH group than in the TD group on d 10, which was consistent with a previous conclusion that the lipid profile is affected negatively in neonatal calves with diarrhea (Bozukluhan et al., 2017). Levels of GH were lower in the RH group than in the TD and RS groups on d 10, which might be because diarrhea affects the release of GH in early postnatal calves (Brückmann et al., 2000). Blood urea nitrogen is normally produced by the rumen and by hepatic deamination of amino acids by the liver (Kim et al., 2021). On d 60, BUN in the TD group was significantly higher than that in the other 2 bedding groups, but ruminal amination-N was the same in the 3 groups; we speculated that calves in the TD group had more hepatic deamination of amino acids by the liver but the reason underlying this speculation needs further study. In summary, TD and RS were better bedding materials than RH in terms of growth performance of suckling buffalo calves.

Calf behavior can reflect the comfort of bedding materials, and some behaviors can influence calves' growth performance. The MC of bedding is related to the lying time of calves: the higher the MC of bedding materials, the shorter calves' lying time (Camiloti et al., 2012). Our data showed no difference in lying time among the 3 bedding groups, despite the increasing MC of bedding materials over time. We speculated that the MC of the bedding materials during the experimental period was within the calves' tolerance. Notably, we found that bedding material eating behavior was affected by bedding type. Although RH and RS are both roughages, RS is a soft and palatable straw that can be considered extra roughage in addition to the starter feed and green hay. Thus, the bedding material eating time in RS group increased with calves' age. However, TD is inedible, and the bedding material eating time in TD remained at a low level, declining to zero at the end of the experiment. The above results indicated no difference in the comfort of the 3 bedding materials for suckling buffalo calves, but bedding type could affect the bedding material eating behavior of calves.

We speculated that bedding material eating behavior would influence rumen fermentation; hence, we

measured rumen fermentation on d 60. Rumen development is the key to growth performance of calves after weaning. The rumen is the main organ digesting and absorbing VFA and ammonia-N in ruminants (Gäbel and Sehested, 1997; Ma et al., 2021). It has been reported that newborn calves rely on the abomasum and small intestine to obtain nutrition and energy (Heinrichs et al., 2005). About 4 wk later, the rumen becomes fully functional and becomes the main organ of digestion (Austin et al., 2018). Adding roughage into calf feed can promote the growth of calf rumen papilla, enrich their rumen microbiota, and increase ruminal acetate (Suárez et al., 2007; Mojahedi et al., 2018). Butyric acid is an important VFA in the rumen and a sign of rumen development (Plöger et al., 2012); it not only promotes the growth of rumen epithelia but also contributes to the growth and maturation of rumen papilla in calves (Mentschel et al., 2001; Górka et al., 2018). Starch and high-sugar feed can promote the production of butyric acid (Lesmeister and Heinrichs, 2005). Our data showed that the butyric acid content in rumen was higher in the TD group than in the RH and RS groups, which might be because that calves bedded with TD ate more starter feed with a high starch content. Calves in the RS group exhibited a longer bedding material eating time, indicating more fiber intake, which explained the higher acetic acid content in the RS group compared with the TD group; this result is consistent with one previous report (Suárez et al., 2007). To summarize, TD bedding could increase the butyric acid content in the calf's rumen, whereas RS bedding could increase the acetate content in the calf's rumen. Further studies are needed to determine the effect of bedding materials on rumen development.

Cost is one of the most important factors that farmers consider when choosing bedding materials (Leso et al., 2020), and a high cost may limit the application of bedding materials (Yajima et al., 2017). In general, dry and green wood materials are the most expensive, followed by sawdust or shavings (Black et al., 2013). In this research, RS cost the most, and RH and TD cost the same. Furthermore, TD provides a strategy for dung treatment, which could save on the cost of disposing of cow dung. To summarize, TD is the most economical because of its low cost and recycling of cow dung.

CONCLUSIONS

Our results showed the feasibility of TD as a bedding for suckling buffalo calves. Compared with RH and RS, we found that TD was superior among the 3 bedding materials with the lowest cost and safest microbial composition, which contributes to the health and growth performance of calves. Thus, TD as a bedding material has potential applications in calf raising.

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